

Lecture 4

Conditional Probability

02.03.2026

Outline

- 1 Conditional probability
- 2 Bayes theorem
- 3 Independent events

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Conditional probability: motivation

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Suppose you were given the extra information that the die roll was an odd number (hence 1, 3 or 5)

Conditional on this new information, the probability of a one is now $1/3$

Conditional probability

Definition

Given two events A and B with $P(B) > 0$, the conditional probability of A given B has occurred is defined as

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- When B is the sample space: $P(A \mid B) = P(A)$
- Intuition: in a sample space with equally likely outcomes,

$$P(A \mid B) = \frac{\#(A \cap B)}{\#(B)}$$

Example

Consider the die roll example: $B = \{1 \text{ or } 3 \text{ or } 5\}$, $A = \{1\}$

$$\begin{aligned} P(\text{get 1 given that roll is odd}) &= P(A \mid B) \\ &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(\text{get 1})}{P(\text{get 1 or 3 or 5})} \\ &= \frac{1/6}{3/6} = \frac{1}{3} \end{aligned}$$

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In context,

$$P(A \cap B) = P(\{(4, 3)\}) = 1/36, \quad P(B) = 1/6$$

$$P(A \mid B) = \frac{1/36}{1/6} = \frac{1}{6}$$

Example: A family has two children. At least one of them is a boy. What is the probability that both children are boys?

Propositions of conditional probability

① $P(A \mid A) = 1$

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$$\textcircled{3} \quad P(A^c \mid B) = 1 - P(A \mid B)$$

Multiplication rule

By the definition of conditional probability, the joint probability of A and B is

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- Usually, $P(A)$ and $P(B)$ are called marginal probabilities.

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Generalize to n events: chaining of probabilities

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1, A_2) \cdots P(A_n \mid A_1, \dots, A_{n-1})$$

A box contains 8 white balls and 4 black ones. 3 balls are drawn at random without replacement.

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- ② What's the probability of getting black in draw 2?

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- ❶ Multiplication rule

$$\begin{aligned} P(W_1B_2B_3) &= P(W_1)P(B_2 | W_1)P(B_3 | W_1B_2) \\ &= \frac{8}{12} \cdot \frac{4}{11} \cdot \frac{3}{10} \end{aligned}$$

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- ❷ If we want to consider draw 1, which can be W_1 or B_1 (disjoint!)

$$\begin{aligned} P(B_2) &= P(W_1B_2) + P(B_1B_2) \\ &= \frac{8}{12} \cdot \frac{4}{11} + \frac{4}{12} \cdot \frac{3}{11} = \frac{44}{132} = \frac{4}{12} \end{aligned}$$

Example: An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.

Law of total probability

For events $A_1, \dots, A_n$ are **disjoint**, and

$$\bigcup_{i=1}^n A_i = S,$$

then for any event B ,

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$$P(B) = P(B \mid A_1)P(A_1) + \dots + P(B \mid A_n)P(A_n)$$

- Such collection of sets $A_1, \dots, A_n$ is also called a partition of sample space.

Example: a firm is considering a drug-testing program for its employees, but before it begins it wants to know the scope of the problem if any exists. Realizing the sensitivity of this issue, the personnel director decides to use a randomized response survey. It is believed that respondents are more likely to be honest when such forms are used. Each employee is asked to flip a fair coin. If the coin comes up heads, answer the question "Do you carpool to work?". If the coin comes up tails, answer the question "Have you used illegal drugs within the last month?". Assume that all employees answer the survey honestly. Out of 8000 responses, 1420 answered "YES". Suppose the firm knows that 35% of its employees carpool to work. What is the probability that an employee chosen at random used illegal drugs within the last month?

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$$P(B) = P(B | A_1)P(A_1) + P(B | A_2)P(A_2)$$

$$P(\text{YES}) = P(\text{carpool})P(\text{head}) + P(\text{drug})P(\text{tail})$$

$$\frac{1420}{8000} = 0.35 \times 0.5 + P(\text{drug}) \times 0.5$$

$$P(\text{drug}) = 0.005$$

Recap

- Marginal probability: $P(A), P(B)$
- Joint probability: $P(A \cap B)$
- Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Recap

- Marginal probability: $P(A), P(B)$
- Joint probability: $P(A \cap B)$
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$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- Multiplication rule:

$$P(A \cap B) = P(A \mid B) \times P(B)$$

- Law of total probability: for a partition $\{A_1, A_2, \dots, A_n\}$ of S ,

$$P(B) = \sum_{j=1}^n P(B \mid A_j)P(A_j)$$

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Bayes theorem (also called Bayes rule)

Suppose events $A_1, \dots, A_n$ are disjoint, and $\bigcup_{i=1}^n A_i = S$, with $P(A_i) > 0$, $i = 1, 2, \dots, n$. Then for any event B with $P(B) > 0$,

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$$\begin{aligned} P(A_i | B) &= \frac{P(B | A_i)P(A_i)}{\sum_{j=1}^n P(B | A_j)P(A_j)}, \quad i = 1, \dots, n \\ &= \frac{P(B | A_i)P(A_i)}{P(B | A_1)P(A_1) + \dots + P(B | A_n)P(A_n)} \end{aligned}$$

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- $P(A_i)$ is often called prior probability
- $P(A_i | B)$ is called posterior probability.

Example: In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and $1 - p$ be the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?

Example: A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

A factory has three machines, M_1 , M_2 and M_3 , that produce items (say, lightbulbs). It is impossible to tell which machine produced a particular item, but some are defective. Here are the known numbers:

machine	proportion of items made	defective item probability
M1	0.2	0.001
M2	0.3	0.002
M3	0.5	0.003

An inspector pick an item, test it, and find it defective. What is the probability that it was made by machine M_2 ?

Recap

Bayes theorem

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_{j=1}^n P(B | A_j)P(A_j)}, \quad i = 1, \dots, n$$

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Product rule for independent events

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Events A and B are independent if

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- If A and B are independent, then

$$P(A \mid B) = \frac{P(A)P(B)}{P(B)} = P(A).$$

Knowing B doesn't affect the odds of A .

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- Method 1: sample space $S = \{HH, TT, HT, TH\}$.

Since one out of four possible outcomes match this definition, the probability is $\frac{1}{4}$.

Inefficient way if the number of trials was much higher.

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- Method 1: sample space $S = \{HH, TT, HT, TH\}$.

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Inefficient way if the number of trials was much higher.

- Method 2: use independence.

$$P(\text{T on the first toss}) \times P(\text{T on the second toss}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Properties of independence

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- Independent is not disjoint.

Mutually independent

- Three events A, B, C are called mutually independent if

1

$$P(A \cap B) = P(A)P(B)$$

$$P(B \cap C) = P(B)P(C)$$

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2

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

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$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

- If only (1) holds but not (2), then A, B, C are called pairwise independent.

Example: pairwise indpt but not mutually indpt

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- Events $A_1, A_2, \dots, A_n$ are called independent if for any $1 \leq r \leq n$ of them

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_r}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_r})$$

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A fair coin is tossed 5 times. Compute $P(5 \text{ tails})$.

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$$P(T_1 T_2 T_3 T_4 T_5) = P(T_1)P(T_2)P(T_3)P(T_4)P(T_5) = \frac{1}{2^5}$$